APT Overlap: APT28 vs APT29 vs Sandworm Team (Parent T-codes)

Technique overlap across 3 selected groups | 129 parent T-codes (sub-techniques rolled up) | campaigns included | software-derived excluded

32 RED techniques
Used by ALL 3 selected groups

31 YELLOW techniques
Used by more than one, but not all

66 LIGHT GREEN techniques
Used by only ONE selected group

Selected Groups

ID	Group	Aliases	T-codes
G0007	APT28	IRON TWILIGHT, SNAKEMACKEREL, Swallowtail, Group 74, Sednit, Sofacy, Pawn Storm, Fancy Bear, STRONTIUM, Tsar Team, Threat Group-4127, TG-4127, Forest Blizzard, FROZENLAKE, GruesomeLarch	77
G0016	APT29	IRON RITUAL, IRON HEMLOCK, NobleBaron, Dark Halo, NOBELIUM, UNC2452, YTTRIUM, The Dukes, Cozy Bear, CozyDuke, SolarStorm, Blue Kitsune, UNC3524, Midnight Blizzard	72
G0034	Sandworm Team	ELECTRUM, Telebots, IRON VIKING, BlackEnergy (Group), Quedagh, Voodoo Bear, IRIDIUM, Seashell Blizzard, FROZENBARENTS, APT44	75

Coverage by Tactic

Tactic	RED	YELLOW	LIGHT GREEN	Total
Reconnaissance	2	2	5	**9**
Resource Development	4	2	1	**7**
Initial Access	5	1	3	**9**
Execution	3	2	4	**9**
Persistence	4	5	6	**15**
Privilege Escalation	2	6	4	**12**
Stealth	6	0	9	**15**
Defense Impairment	1	2	3	**6**
Credential Access	2	5	7	**14**
Discovery	1	5	7	**13**
Lateral Movement	1	1	4	**6**
Collection	2	4	5	**11**
Command and Control	4	2	8	**14**
Exfiltration	0	1	3	**4**
Impact	0	1	7	**8**

Technique Matrix

parent T-codes only (sub-technique use rolled up into the parent)

RED (32) YELLOW (31) LIGHT GREEN (66)

Reconnaissance 9 techniques	Resource Development 7 techniques	Initial Access 9 techniques	Execution 9 techniques	Persistence 15 techniques	Privilege Escalation 12 techniques	Stealth 15 techniques	Defense Impairment 6 techniques	Credential Access 14 techniques	Discovery 13 techniques	Lateral Movement 6 techniques	Collection 11 techniques	Command and Control 14 techniques	Exfiltration 4 techniques	Impact 8 techniques
T1589 3/3 Gather Victim Identity Information	T1583 3/3 Acquire Infrastructure	T1078 3/3 Valid Accounts	T1059 3/3 Command and Scripting Interpreter	T1078 3/3 Valid Accounts	T1078 3/3 Valid Accounts	T1027 3/3 Obfuscated Files or Information	T1685 3/3 Disable or Modify Tools	T1003 3/3 OS Credential Dumping	T1083 3/3 File and Directory Discovery	T1021 3/3 Remote Services	T1005 3/3 Data from Local System	T1071 3/3 Application Layer Protocol	T1048 2/3 Exfiltration Over Alternative Protocol	T1561 2/3 Disk Wipe
T1595 3/3 Active Scanning	T1584 3/3 Compromise Infrastructure	T1133 3/3 External Remote Services	T1203 3/3 Exploitation for Client Execution	T1098 3/3 Account Manipulation	T1098 3/3 Account Manipulation	T1036 3/3 Masquerading	T1484 2/3 Domain or Tenant Policy Modification	T1110 3/3 Brute Force	T1016 2/3 System Network Configuration Discovery	T1550 2/3 Use Alternate Authentication Material	T1213 3/3 Data from Information Repositories	T1090 3/3 Proxy	T1030 1/3 Data Transfer Size Limits	T1485 1/3 Data Destruction
T1591 2/3 Gather Victim Org Information	T1586 3/3 Compromise Accounts	T1190 3/3 Exploit Public-Facing Application	T1204 3/3 User Execution	T1133 3/3 External Remote Services	T1037 2/3 Boot or Logon Initialization Scripts	T1070 3/3 Indicator Removal	T1686 2/3 Disable or Modify System Firewall	T1040 2/3 Network Sniffing	T1018 2/3 Remote System Discovery	T1072 1/3 Software Deployment Tools	T1056 2/3 Input Capture	T1102 3/3 Web Service	T1041 1/3 Exfiltration Over C2 Channel	T1486 1/3 Data Encrypted for Impact
T1598 2/3 Phishing for Information	T1588 3/3 Obtain Capabilities	T1199 3/3 Trusted Relationship	T1047 2/3 Windows Management Instrumentation	T1505 3/3 Server Software Component	T1053 2/3 Scheduled Task/Job	T1078 3/3 Valid Accounts	T1112 1/3 Modify Registry	T1056 2/3 Input Capture	T1040 2/3 Network Sniffing	T1091 1/3 Replication Through Removable Media	T1074 2/3 Data Staged	T1105 3/3 Ingress Tool Transfer	T1567 1/3 Exfiltration Over Web Service	T1489 1/3 Service Stop
T1590 1/3 Gather Victim Network Information	T1585 2/3 Establish Accounts	T1566 3/3 Phishing	T1053 2/3 Scheduled Task/Job	T1037 2/3 Boot or Logon Initialization Scripts	T1068 2/3 Exploitation for Privilege Escalation	T1140 3/3 Deobfuscate/Decode Files or Information	T1553 1/3 Subvert Trust Controls	T1528 2/3 Steal Application Access Token	T1057 2/3 Process Discovery	T1210 1/3 Exploitation of Remote Services	T1114 2/3 Email Collection	T1001 2/3 Data Obfuscation		T1490 1/3 Inhibit System Recovery
T1592 1/3 Gather Victim Host Information	T1587 2/3 Develop Capabilities	T1195 2/3 Supply Chain Compromise	T1072 1/3 Software Deployment Tools	T1053 2/3 Scheduled Task/Job	T1484 2/3 Domain or Tenant Policy Modification	T1218 3/3 System Binary Proxy Execution	T1556 1/3 Modify Authentication Process	T1539 2/3 Steal Web Session Cookie	T1087 2/3 Account Discovery	T1570 1/3 Lateral Tool Transfer	T1560 2/3 Archive Collected Data	T1573 2/3 Encrypted Channel		T1491 1/3 Defacement
T1593 1/3 Search Open Websites/Domains	T1608 1/3 Stage Capabilities	T1091 1/3 Replication Through Removable Media	T1106 1/3 Native API	T1136 2/3 Create Account	T1546 2/3 Event Triggered Execution	T1006 1/3 Direct Volume Access		T1555 2/3 Credentials from Password Stores	T1033 1/3 System Owner/User Discovery		T1025 1/3 Data from Removable Media	T1092 1/3 Communication Through Removable ...		T1498 1/3 Network Denial of Service
T1594 1/3 Search Victim-Owned Websites		T1189 1/3 Drive-by Compromise	T1559 1/3 Inter-Process Communication	T1546 2/3 Event Triggered Execution	T1547 2/3 Boot or Logon Autostart Execution	T1014 1/3 Rootkit		T1552 1/3 Unsecured Credentials	T1049 1/3 System Network Connections Discovery		T1039 1/3 Data from Network Shared Drive	T1095 1/3 Non-Application Layer Protocol		T1499 1/3 Endpoint Denial of Service
T1596 1/3 Search Open Technical Databases		T1669 1/3 Wi-Fi Networks	T1651 1/3 Cloud Administration Command	T1547 2/3 Boot or Logon Autostart Execution	T1055 1/3 Process Injection	T1055 1/3 Process Injection		T1556 1/3 Modify Authentication Process	T1069 1/3 Permission Groups Discovery		T1113 1/3 Screen Capture	T1132 1/3 Data Encoding		
				T1112 1/3 Modify Registry	T1134 1/3 Access Token Manipulation	T1134 1/3 Access Token Manipulation		T1557 1/3 Adversary-in-the-Middle	T1082 1/3 System Information Discovery		T1119 1/3 Automated Collection	T1219 1/3 Remote Access Tools		
				T1137 1/3 Office Application Startup	T1543 1/3 Create or Modify System Process	T1211 1/3 Exploitation for Stealth		T1558 1/3 Steal or Forge Kerberos Tickets	T1120 1/3 Peripheral Device Discovery		T1557 1/3 Adversary-in-the-Middle	T1568 1/3 Dynamic Resolution		
				T1542 1/3 Pre-OS Boot	T1548 1/3 Abuse Elevation Control Mechanism	T1221 1/3 Template Injection		T1606 1/3 Forge Web Credentials	T1482 1/3 Domain Trust Discovery			T1571 1/3 Non-Standard Port		
				T1543 1/3 Create or Modify System Process		T1542 1/3 Pre-OS Boot		T1621 1/3 Multi-Factor Authentication Reque...	T1680 1/3 Local Storage Discovery			T1572 1/3 Protocol Tunneling		
				T1554 1/3 Compromise Host Software Binary		T1564 1/3 Hide Artifacts		T1649 1/3 Steal or Forge Authentication Certific...				T1665 1/3 Hide Infrastructure		
				T1556 1/3 Modify Authentication Process		T1684 1/3 Social Engineering								

Technique Detail

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Sandworm Team
RED	3/3	T1003	OS Credential Dumping	Credential Access	✔	✔	✔
RED	3/3	T1005	Data from Local System	Collection	✔	✔	✔
RED	3/3	T1021	Remote Services	Lateral Movement	✔	✔	✔
RED	3/3	T1027	Obfuscated Files or Information	Stealth	✔	✔	✔
RED	3/3	T1036	Masquerading	Stealth	✔	✔	✔
RED	3/3	T1059	Command and Scripting Interpreter	Execution	✔	✔	✔
RED	3/3	T1070	Indicator Removal	Stealth	✔	✔	✔
RED	3/3	T1071	Application Layer Protocol	Command and Control	✔	✔	✔
RED	3/3	T1078	Valid Accounts	Initial Access, Persistence, Privilege Escalation, Stealth	✔	✔	✔
RED	3/3	T1083	File and Directory Discovery	Discovery	✔	✔	✔
RED	3/3	T1090	Proxy	Command and Control	✔	✔	✔
RED	3/3	T1098	Account Manipulation	Persistence, Privilege Escalation	✔	✔	✔
RED	3/3	T1102	Web Service	Command and Control	✔	✔	✔
RED	3/3	T1105	Ingress Tool Transfer	Command and Control	✔	✔	✔
RED	3/3	T1110	Brute Force	Credential Access	✔	✔	✔
RED	3/3	T1133	External Remote Services	Initial Access, Persistence	✔	✔	✔
RED	3/3	T1140	Deobfuscate/Decode Files or Information	Stealth	✔	✔	✔
RED	3/3	T1190	Exploit Public-Facing Application	Initial Access	✔	✔	✔
RED	3/3	T1199	Trusted Relationship	Initial Access	✔	✔	✔
RED	3/3	T1203	Exploitation for Client Execution	Execution	✔	✔	✔
RED	3/3	T1204	User Execution	Execution	✔	✔	✔
RED	3/3	T1213	Data from Information Repositories	Collection	✔	✔	✔
RED	3/3	T1218	System Binary Proxy Execution	Stealth	✔	✔	✔
RED	3/3	T1505	Server Software Component	Persistence	✔	✔	✔
RED	3/3	T1566	Phishing	Initial Access	✔	✔	✔
RED	3/3	T1583	Acquire Infrastructure	Resource Development	✔	✔	✔
RED	3/3	T1584	Compromise Infrastructure	Resource Development	✔	✔	✔
RED	3/3	T1586	Compromise Accounts	Resource Development	✔	✔	✔
RED	3/3	T1588	Obtain Capabilities	Resource Development	✔	✔	✔
RED	3/3	T1589	Gather Victim Identity Information	Reconnaissance	✔	✔	✔
RED	3/3	T1595	Active Scanning	Reconnaissance	✔	✔	✔
RED	3/3	T1685	Disable or Modify Tools	Defense Impairment	✔	✔	✔
YELLOW	2/3	T1001	Data Obfuscation	Command and Control	✔	✔	
YELLOW	2/3	T1016	System Network Configuration Discovery	Discovery	✔	✔	
YELLOW	2/3	T1018	Remote System Discovery	Discovery		✔	✔
YELLOW	2/3	T1037	Boot or Logon Initialization Scripts	Persistence, Privilege Escalation	✔	✔	
YELLOW	2/3	T1040	Network Sniffing	Credential Access, Discovery	✔		✔

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Sandworm Team
YELLOW	2/3	T1047	Windows Management Instrumentation	Execution		✔	✔
YELLOW	2/3	T1048	Exfiltration Over Alternative Protocol	Exfiltration	✔	✔	
YELLOW	2/3	T1053	Scheduled Task/Job	Execution, Persistence, Privilege Escalation		✔	✔
YELLOW	2/3	T1056	Input Capture	Credential Access, Collection	✔		✔
YELLOW	2/3	T1057	Process Discovery	Discovery	✔	✔	
YELLOW	2/3	T1068	Exploitation for Privilege Escalation	Privilege Escalation	✔	✔	
YELLOW	2/3	T1074	Data Staged	Collection	✔	✔	
YELLOW	2/3	T1087	Account Discovery	Discovery		✔	✔
YELLOW	2/3	T1114	Email Collection	Collection	✔	✔	
YELLOW	2/3	T1136	Create Account	Persistence		✔	✔
YELLOW	2/3	T1195	Supply Chain Compromise	Initial Access		✔	✔
YELLOW	2/3	T1484	Domain or Tenant Policy Modification	Privilege Escalation, Defense Impairment		✔	✔
YELLOW	2/3	T1528	Steal Application Access Token	Credential Access	✔	✔	
YELLOW	2/3	T1539	Steal Web Session Cookie	Credential Access		✔	✔
YELLOW	2/3	T1546	Event Triggered Execution	Persistence, Privilege Escalation	✔	✔	
YELLOW	2/3	T1547	Boot or Logon Autostart Execution	Persistence, Privilege Escalation	✔	✔	
YELLOW	2/3	T1550	Use Alternate Authentication Material	Lateral Movement	✔	✔	
YELLOW	2/3	T1555	Credentials from Password Stores	Credential Access		✔	✔
YELLOW	2/3	T1560	Archive Collected Data	Collection	✔	✔	
YELLOW	2/3	T1561	Disk Wipe	Impact	✔		✔
YELLOW	2/3	T1573	Encrypted Channel	Command and Control	✔	✔	
YELLOW	2/3	T1585	Establish Accounts	Resource Development		✔	✔
YELLOW	2/3	T1587	Develop Capabilities	Resource Development		✔	✔
YELLOW	2/3	T1591	Gather Victim Org Information	Reconnaissance	✔		✔
YELLOW	2/3	T1598	Phishing for Information	Reconnaissance	✔		✔
YELLOW	2/3	T1686	Disable or Modify System Firewall	Defense Impairment	✔	✔	
LIGHT GREEN	1/3	T1006	Direct Volume Access	Stealth	✔		
LIGHT GREEN	1/3	T1014	Rootkit	Stealth	✔		
LIGHT GREEN	1/3	T1025	Data from Removable Media	Collection	✔		
LIGHT GREEN	1/3	T1030	Data Transfer Size Limits	Exfiltration	✔		
LIGHT GREEN	1/3	T1033	System Owner/User Discovery	Discovery			✔
LIGHT GREEN	1/3	T1039	Data from Network Shared Drive	Collection	✔		
LIGHT GREEN	1/3	T1041	Exfiltration Over C2 Channel	Exfiltration			✔
LIGHT GREEN	1/3	T1049	System Network Connections Discovery	Discovery			✔
LIGHT GREEN	1/3	T1055	Process Injection	Privilege Escalation, Stealth			✔
LIGHT GREEN	1/3	T1069	Permission Groups Discovery	Discovery		✔	
LIGHT GREEN	1/3	T1072	Software Deployment Tools	Execution, Lateral Movement			✔

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Sandworm Team
LIGHT GREEN	1/3	T1082	System Information Discovery	Discovery			✔
LIGHT GREEN	1/3	T1091	Replication Through Removable Media	Initial Access, Lateral Movement	✔		
LIGHT GREEN	1/3	T1092	Communication Through Removable Media	Command and Control	✔		
LIGHT GREEN	1/3	T1095	Non-Application Layer Protocol	Command and Control			✔
LIGHT GREEN	1/3	T1106	Native API	Execution			✔
LIGHT GREEN	1/3	T1112	Modify Registry	Persistence, Defense Impairment			✔
LIGHT GREEN	1/3	T1113	Screen Capture	Collection	✔		
LIGHT GREEN	1/3	T1119	Automated Collection	Collection	✔		
LIGHT GREEN	1/3	T1120	Peripheral Device Discovery	Discovery	✔		
LIGHT GREEN	1/3	T1132	Data Encoding	Command and Control			✔
LIGHT GREEN	1/3	T1134	Access Token Manipulation	Privilege Escalation, Stealth	✔		
LIGHT GREEN	1/3	T1137	Office Application Startup	Persistence	✔		
LIGHT GREEN	1/3	T1189	Drive-by Compromise	Initial Access	✔		
LIGHT GREEN	1/3	T1210	Exploitation of Remote Services	Lateral Movement	✔		
LIGHT GREEN	1/3	T1211	Exploitation for Stealth	Stealth	✔		
LIGHT GREEN	1/3	T1219	Remote Access Tools	Command and Control			✔
LIGHT GREEN	1/3	T1221	Template Injection	Stealth	✔		
LIGHT GREEN	1/3	T1482	Domain Trust Discovery	Discovery		✔	
LIGHT GREEN	1/3	T1485	Data Destruction	Impact			✔
LIGHT GREEN	1/3	T1486	Data Encrypted for Impact	Impact			✔
LIGHT GREEN	1/3	T1489	Service Stop	Impact			✔
LIGHT GREEN	1/3	T1490	Inhibit System Recovery	Impact			✔
LIGHT GREEN	1/3	T1491	Defacement	Impact			✔
LIGHT GREEN	1/3	T1498	Network Denial of Service	Impact	✔		
LIGHT GREEN	1/3	T1499	Endpoint Denial of Service	Impact			✔
LIGHT GREEN	1/3	T1542	Pre-OS Boot	Persistence, Stealth	✔		
LIGHT GREEN	1/3	T1543	Create or Modify System Process	Persistence, Privilege Escalation			✔
LIGHT GREEN	1/3	T1548	Abuse Elevation Control Mechanism	Privilege Escalation		✔	
LIGHT GREEN	1/3	T1552	Unsecured Credentials	Credential Access		✔	
LIGHT GREEN	1/3	T1553	Subvert Trust Controls	Defense Impairment		✔	
LIGHT GREEN	1/3	T1554	Compromise Host Software Binary	Persistence			✔
LIGHT GREEN	1/3	T1556	Modify Authentication Process	Persistence, Defense Impairment, Credential Access		✔	
LIGHT GREEN	1/3	T1557	Adversary-in-the-Middle	Credential Access, Collection	✔		
LIGHT GREEN	1/3	T1558	Steal or Forge Kerberos Tickets	Credential Access		✔	
LIGHT GREEN	1/3	T1559	Inter-Process Communication	Execution	✔		
LIGHT GREEN	1/3	T1564	Hide Artifacts	Stealth	✔		

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Sandworm Team
LIGHT GREEN	1/3	T1567	Exfiltration Over Web Service	Exfiltration	✔		
LIGHT GREEN	1/3	T1568	Dynamic Resolution	Command and Control		✔	
LIGHT GREEN	1/3	T1570	Lateral Tool Transfer	Lateral Movement			✔
LIGHT GREEN	1/3	T1571	Non-Standard Port	Command and Control			✔
LIGHT GREEN	1/3	T1572	Protocol Tunneling	Command and Control			✔
LIGHT GREEN	1/3	T1590	Gather Victim Network Information	Reconnaissance			✔
LIGHT GREEN	1/3	T1592	Gather Victim Host Information	Reconnaissance			✔
LIGHT GREEN	1/3	T1593	Search Open Websites/Domains	Reconnaissance			✔
LIGHT GREEN	1/3	T1594	Search Victim-Owned Websites	Reconnaissance			✔
LIGHT GREEN	1/3	T1596	Search Open Technical Databases	Reconnaissance	✔		
LIGHT GREEN	1/3	T1606	Forge Web Credentials	Credential Access		✔	
LIGHT GREEN	1/3	T1608	Stage Capabilities	Resource Development			✔
LIGHT GREEN	1/3	T1621	Multi-Factor Authentication Request Generation	Credential Access		✔	
LIGHT GREEN	1/3	T1649	Steal or Forge Authentication Certificates	Credential Access		✔	
LIGHT GREEN	1/3	T1651	Cloud Administration Command	Execution		✔	
LIGHT GREEN	1/3	T1665	Hide Infrastructure	Command and Control		✔	
LIGHT GREEN	1/3	T1669	Wi-Fi Networks	Initial Access	✔		
LIGHT GREEN	1/3	T1680	Local Storage Discovery	Discovery		✔	
LIGHT GREEN	1/3	T1684	Social Engineering	Stealth	✔		